

Sergio Bampi

professor

基本信息

姓名: Sergio Bampi

所属院校: 南里奥格兰德联邦大学

地区: Brazil

学科领域

集成电路设计与集成系统

电子与计算机工程

微电子科学与工程

数字媒体技术

研究领域

低功耗电路与能量管理

数字集成电路设计

低功耗设计与能耗优化

模拟与数模混合电路

数字音视频处理

嵌入式系统与硬件设计

超大规模集成电路设计

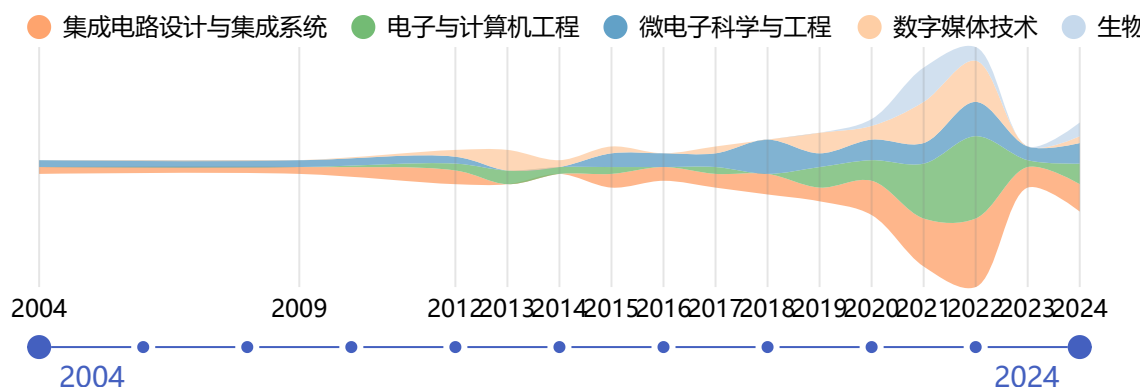
计算机体系结构

EDA工具与电路验证

个人简介

Sergio Bampi is a professor at the Informatics Institute of the Federal University of Rio Grande do Sul. His research interests include integrated circuits design, digital low power CMOS circuits, and architectures and algorithms for video/image processing. Since 2020, he has been cited 2222 times with an h-index of 22 and an i10-index of 66. He has co-authored numerous papers with scholars from various institutions, such as Federal University of Pelotas, Federal University of Rio Grande do Sul, and Karlsruhe Institute of Technology.

近年发表的论文



On the Efficacy and Vulnerabilities of Logic Locking in Tree-Based Machine Learning

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

de Abreu, Brunno Alves; Paim, Guilherme; Alrahis, Lilas; Flores, Paulo; Sinanoglu, Ozgur; Bampi, Sergio; Amrouch, Hussam

被引用次数: 0

2024

VLSI Architectures of Approximate Arithmetic Units Applied to Parallel Sensors Calibration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

da Rosa, Morgana Macedo Azevedo; da Costa, Patricia Ucker Leleu; da Costa, Eduardo Antonio Cesar; Soares, Rafael I.; Bampi, Sergio

被引用次数: 4

2024

VLSI Architecture for Energy-Efficient and Accurate Pre-Processing Pan-Tompkins Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ribeiro, Leo; da Costa, Patricia; Paim, Guilherme; da Costa, Eduardo A.; de Almeida, Sergio Jose Melo; Bampi, Sergio

被引用次数: 2

2024

ReAdapt-II: Energy-Quality Optimizations for VLSI Adaptive Filters Through Automatic Reconfiguration and Built-In Iterative Dividers

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Pereira, Pedro T. L.; Costa, Patricia Ucker L.; da Costa, Eduardo; Flores, Paulo; Bampi, Sergio

被引用次数: 0

2024

The Digital-Based Operational Transconductance Amplifier: Evolution and Perspective

2024 37TH SBC/SBMICRO/IEEE SYMPOSIUM ON INTEGRATED CIRCUITS AND SYSTEMS DESIGN, SBCCI 2024

Toledo, Pedro; Bampi, Sergio; Crovetto, Paolo

被引用次数: 0

2024

Statistical Analysis of VVC Residual and Entropy Coding aiming Efficient Hardware Design

15TH IEEE LATIN AMERICAN SYMPOSIUM ON CIRCUITS AND SYSTEMS, LASCAS 2024

Cardoso, Gabriel Bitencourt; Tomm, Daniel Felipe; Gomes, Jiovana Sousa; Wuerdig, Rodrigo Nogueira; Bampi, Sergio; Livi Ramos, Fabio Luis

被引用次数: 0

2024

Alternative Reference Samples to Improve Coding Efficiency for Parallel Intra Prediction Solutions

15TH IEEE LATIN AMERICAN SYMPOSIUM ON CIRCUITS AND SYSTEMS, LASCAS 2024

Storch, Iago; Roma, Nuno; Palomino, Daniel; Bampi, Sergio

被引用次数: 0

2024

ReAdapt: A Reconfigurable Datapath for Runtime Energy-Quality Scalable Adaptive Filters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lopes Pereira, Pedro Taua; Paim, Guilherme; Cesar da Costa, Eduardo Antonio; Melo de Almeida, Sergio Jose; Bampi, Sergio

被引用次数: 3

2023

Robustness Analysis of 3-2 Adder Compressor Designed in 7-nm FinFET Technology

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Andrade, Gerson; Silva, Matheus; Schneider, Cinthia; Paim, Guilherme; Bampi, Sergio; Costa, Eduardo; Zimpeck, Alexandra

被引用次数: 2

2023

Compact CMOS-Compatible Majority Gate Using Body Biasing in FDSOI Technology

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

de Abreu, Brunno Alves; Mema, Albi; Thomann, Simon; Paim, Guilherme; Flores, Paulo; Bampi, Sergio; Amrouch, Hussam

被引用次数: 2

2023

AxPPA: Approximate Parallel Prefix Adders

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Azevedo da Rosa, Morgana Macedo; Paim, Guilherme; Leleu da Costa, Patricia Ucker; Cesar da Costa, Eduardo Antonio; Soares, Rafael, I; Bampi, Sergio

被引用次数: 18

2023

A framework for designing power-efficient inference accelerators in tree-based learning applications

ENGINEERING APPLICATIONS OF ARTIFICIAL INTELLIGENCE

Abreu, Brunno; Grellert, Mateus; Bampi, Sergio

被引用次数: 7

2022

A Framework for Crossing Temperature-Induced Timing Errors Underlying Hardware Accelerators to the Algorithm and Application Layers

IEEE TRANSACTIONS ON COMPUTERS

Paim, Guilherme; Amrouch, Hussam; Rocha, Leandro M. G.; Abreu, Brunno; Cesar da Costa, Eduardo Antonio; Bampi, Sergio; Henkel, Joerg

被引用次数: 4

2022

Low-Voltage,Low-Area, nW-Power CMOS Digital-Based Biosignal Amplifier

IEEE ACCESS

Toledo, Pedro; Crovetti, Paolo S.; Klimach, Hamilton D.; Musolino, Francesco; Bampi, Sergio

被引用次数: 8

2022

FastInter360: A Fast Inter Mode Decision for HEVC 360 Video Coding

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Storch, Iago; Agostini, Luciano; Zatt, Bruno; Bampi, Sergio; Palomino, Daniel

被引用次数: 9

2022

Bridging the Gap Between Voltage Over-Scaling and Joint Hardware Accelerator-Algorithm Closed-Loop

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Paim, Guilherme; Amrouch, Hussam; da Costa, Eduardo Antonio Cesar; Bampi, Sergio; Henkel, Joerg

被引用次数: 5

2022

Energy-Quality Scalable Design Space Exploration of Approximate FFT Hardware Architectures

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Pereira, Pedro Tavares Lopes; da Costa, Patricia Ucker Leleu; Ferreira, Guilherme da Costa; de Abreu, Brunno Alves; Paim, Guilherme; da Costa, Eduardo Antonio Cesar; Bampi, Sergio

被引用次数: 10

2022

C2PAx: Complexity-Aware Constant Parameter Approximation for Energy-Efficient Tree-Based Machine Learning Accelerators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

de Abreu, Brunno Alves; Paim, Guilherme; Grellert, Mateus; Bampi, Sergio

被引用次数: 10

2022

Design of High-IF Discrete-Time Receivers for IoT: Demystifying Aliasing Trade-Offs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ferreira, Sandro Binsfeld; Baumgratz, Filipe D.; Bampi, Sergio; Staszewski, Robert Bogdan

被引用次数: 1

2022

The 4-2 Fused Adder-Subtractor Compressor for Low-Power Butterfly-Based Hardware Architectures

CIRCUITS SYSTEMS AND SIGNAL PROCESSING

Silveira, Bianca; Paim, Guilherme; Abreu, Brunno Alves; Ferreira, Rafael dos Santos; Diniz, Claudio Machado; Cesar da Costa, Eduardo Antonio; Bampi, Sergio

被引用次数: 1

2022

Low-Power High-Throughput Architecture for AV1 Arithmetic Decoder

IEEE DESIGN & TEST

Gomes, Jiovana Sousa; Bampi, Sergio; Bitencourt, Tulio Pereira; Livi Ramos, Fabio Luis

被引用次数: 0

2022

Power-Saving 8K Real-Time AV1 Arithmetic Encoder Architecture

IEEE DESIGN & TEST

Bitencourt, Tulio Pereira; Livi Ramos, Fabio Luis; Bampi, Sergio

Design of a low power and robust VLSI power line interference canceler with optimized arithmetic operators

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

da Rosa, Morgana Macedo Azevedo; da Costa, Patricia Ucker; da Costa, Eduardo Antonio Cesar; Almeida, Sergio J. M.; Paim, Guilherme; Bampi, Sergio

被引用次数: 2

2022

AppGNN: Approximation-Aware Functional Reverse Engineering using Graph Neural Networks

2022 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Buecher, Tim; Alrahis, Lilas; Paim, Guilherme; Bampi, Sergio; Sinanoglu, Ozgur; Amrouch, Hussam

被引用次数: 4

2022

Improved Approximate Multipliers for Single-Precision Floating-Point Hardware Design

2022 IEEE 13TH LATIN AMERICAN SYMPOSIUM ON CIRCUITS AND SYSTEMS (LASCAS)

da Costa, Patricia; Pereira, Pedro T. L.; Abreu, Brunno A.; Paim, Guilherme; da Costa, Eduardo; Bampi, Sergio

被引用次数: 2

2022

On the Netlist Gate-level Pruning for Tree-based Machine Learning Accelerators

2022 IEEE 13TH LATIN AMERICAN SYMPOSIUM ON CIRCUITS AND SYSTEMS (LASCAS)

de Abreu, Brunno A.; Paim, Guilherme; Castro-Godinez, Jorge; Grellert, Mateus; Bampi, Sergio

被引用次数: 0

2022

Hybrid Comparator and Window Switching Scheme for low-power SAR ADC

2022 IEEE 13TH LATIN AMERICAN SYMPOSIUM ON CIRCUITS AND SYSTEMS (LASCAS)

Canal, Bruno; Klimach, Hamilton; Bampi, Sergio; Balen, Tiago R.

被引用次数: 0

2022

Video Decoder Improvements with Near-Data Speculative Motion Compensation Processing

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

de Souza, Garrenlus; Azambuja, Jose Rodrigo; Zatt, Bruno; Zanata, Marco A.; Bampi, Sergio; Sampaio, Felipe

被引用次数: 0

2022

GPU-Acceleration of Affine Prediction in the Versatile Video Coding

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Storch, Iago; Palomino, Daniel; Bampi, Sergio

被引用次数: 1

2022

Mitigating Asynchronous QDI Drawbacks on MAC Operators with Approximate Multipliers

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Wuerdig, Rodrigo N.; Sartori, Marcos L. L.; Abreu, Brunno A.; Bampi, Sergio; Calazans, Ney L. V.

被引用次数: 1

2022

AxRSU: Approximate Radix-4 Squarer Unit

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

da Rosa, Morgana M. A.; Paim, Guilherme; Castro-Godinez, Jorge; da Costa, Eduardo A. C.; Soares, Rafael I.; Bampi, Sergio

被引用次数: 4

2022

Area and Power Efficient 8K Real-Time Design for AV1 Arithmetic Decoding

2022 PICTURE CODING SYMPOSIUM (PCS)

Gomes, Jiovana Sousa; Bitencourt, Tulio Pereira; Bampi, Sergio; Livi Ramos, Fabio Luis

被引用次数: 3

2022

Power-Throughput Trade-off Analysis for a Novel Multi-Boolean AV1 Arithmetic Encoder Design

2022 PICTURE CODING SYMPOSIUM (PCS)

Bitencourt, Tulio Pereira; Livi Ramosy, Fabio Luis; Bampi, Sergio

被引用次数: 2

2022

An Efficient Exponential Unit Designed in VLSI CMOS with Custom Operators

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

da Costa, Patricia; da Rosa, Morgana; Paim, Guilherme; da Costa, Eduardo; Soares, Rafael; Bampi, Sergio

被引用次数: 2

2022

Discrete Haar Wavelet Transform Hardware Design for Energy-Efficient Image Watermarking

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

da Rosat, Morgana; Paim, Guilherme; Soares, Rafael; da Costa, Eduardo; Bampi, Sergio

被引用次数: 3

2022

Exploring Approximate Arithmetic Units for a Power-Efficient Kalman Gain VLSI Design

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Pereira, Pedro T. L.; Paim, Guilherme; da Costa, Eduardo; Almeida, Sergio J. M.; Bampi, Sergio

被引用次数: 1

2022

Low-energy motion estimation memory system with dynamic management

JOURNAL OF REAL-TIME IMAGE PROCESSING

Silveira, Dieison Soares; Amaral, Livia; Povala, Guilherme; Zatt, Bruno; Agostini, Luciano Volcan; Porto, Marcelo Schiavon; Bampi, Sergio

被引用次数: 1

2021

Coding mode decision algorithm for fast HEVC transrating using heuristics and machine learning

JOURNAL OF REAL-TIME IMAGE PROCESSING

Grellert, Mateus; da Silva Cruz, Luis A.; Zatt, Bruno; Bampi, Sergio

被引用次数: 3

2021

Low-power fast Fourier transform hardware architecture combining a split-radix butterfly and efficient adder compressors

IET COMPUTERS AND DIGITAL TECHNIQUES

Ferreira, Guilherme; Paim, Guilherme; Rocha, Leandro M. G.; Santana, Gustavo M.; Neuenfeld, Renato H.; Costa, Eduardo A. C.; Bampi, Sergio

被引用次数: 7

2021

Energy-Throughput Configurable Design for Video Processing Binary Arithmetic Encoder

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Livi Ramos, Fabio Luis; Zatt, Bruno; Porto, Marcelo; Bampi, Sergio

被引用次数: 6

2021

Approximate Pruned and Truncated Haar Discrete Wavelet Transform VLSI Hardware for Energy-Efficient ECG Signal Processing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Seidel, Henrique Bestani; Azevedo da Rosa, Morgana Macedo; Paim, Guilherme; Cesar da Costa, Eduardo Antonio; Almeida, Sergio J. M.; Bampi, Sergio

被引用次数: 34

2021

On the Resiliency of NCFET Circuits Against Voltage Over-Scaling

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Paim, Guilherme; Zervakis, Georgios; Pahwa, Girish; Chauhan, Yogesh Singh; Cesar da Costa, Eduardo Antonio; Bampi, Sergio; Henkel, Joerg; Amrouch, Hussam

被引用次数: 15

2021

A 300mV-Supply, Sub-nW-Power Digital-Based Operational Transconductance Amplifier

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Toledo, Pedro; Crovetto, Paolo; Klimach, Hamilton; Bampi, Sergio; Aiello, Orazio; Alioto, Massimo

被引用次数: 17

2021

An Energy-Efficient Haar Wavelet Transform Architecture for Respiratory Signal Processing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

da Rosa, Morgana M.; Seidel, Henrique B.; Paim, Guilherme; da Costa, Eduardo A. C.; Almeida, Sergio; Bampi, Sergio

被引用次数: 5

2021

Architectural Exploration for Energy-Efficient Fixed-Point Kalman Filter VLSI Design

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lopes Pereira, Pedro Taua; Paim, Guilherme; Leleu da Costa, Patricia Ucker; Cesar da Costa, Eduardo Antonio; Melo de Almeida, Sergio Jose; Bampi, Sergio

被引用次数: 6

2021

Exploring NLMS-Based Adaptive Filter Hardware Architectures for Eliminating Power Line Interference in EEG Signals

CIRCUITS SYSTEMS AND SIGNAL PROCESSING

La Rosa, Andrei B.; Pereira, Pedro T. L.; Ucker, Patricia; Paim, Guilherme; da Costa, Eduardo A. C.; Bampi, Sergio; Almeida, Sergio

被引用次数: 9

2021

Low-voltage dynamic comparator using positive feedback bulk effect on a floating inverter amplifier

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Canal, Bruno; Klimach, Hamilton D.; Bampi, Sergio; Balen, Tiago R.

被引用次数: 5

2021

A new feedback strategy to boost Q-factor of charge-sharing bandpass filters

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Baumgratz, Filipe D.; Ferreira, Sandro B.; Steyaert, Michiel S. J.; Tavernier, Filip; Bampi, Sergio

被引用次数: 1

2021

Analysis of ultrasonic vocalizations from mice using computer vision and machine learning

ELIFE

Fonseca, Antonio H. O.; Santana, Gustavo M.; Ortiz, Gabriela M. Bosque; Bampi, Sergio; Dietrich, Marcelo O.

被引用次数: 39

2021

Fixed-Point NLMS and IPNLMS VLSI Architectures for Accurate FECG and FHR Processing

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

da Costa, Patricia Ucker; Paim, Guilherme; Giacomini Rocha, Leandro Mateus; Cesar da Costa, Eduardo Antonio; Melo de Almeida, Sergio Jose; Bampi, Sergio

被引用次数: 13

2021

QUALITY AND COMPLEXITY ASSESSMENT OF LEARNING-BASED IMAGE COMPRESSION SOLUTIONS

2021 IEEE INTERNATIONAL CONFERENCE ON IMAGE PROCESSING (ICIP)

Dick, Joao; Abreu, Brunno; Grellert, Mateus; Bampi, Sergio

被引用次数: 1

2021

A Power-Efficient FFT Hardware Architecture Exploiting Approximate Adders

2021 IEEE 12TH LATIN AMERICA SYMPOSIUM ON CIRCUITS AND SYSTEM (LASCAS)

Ferreira, Guilherme; Pereira, Pedro T. L.; Paim, Guilherme; Costa, Eduardo; Bampi, Sergio

被引用次数: 1

2021

Boosting the Efficiency of the Harmonics Elimination VLSI Architecture by Arithmetic Approximations

2021 28TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS, AND SYSTEMS (IEEE ICECS 2021)

da Costa, Patricia; Pereira, Pedro T. L.; Paim, Guilherme; da Costa, Eduardo; Bampi, Sergio

被引用次数: 0

2021

LSTM-only Model for Low-complexity HR Estimation from Wrist PPG

2021 43RD ANNUAL INTERNATIONAL CONFERENCE OF THE IEEE ENGINEERING IN MEDICINE & BIOLOGY SOCIETY (EMBC)

Rocha, Leandro Giacomini; Paim, Guilherme; Biswas, Dwaipayan; Bampi, Sergio; Catthoor, Francky; Van Hoof, Chris; Van Helleputte, Nick

被引用次数: 5

2021

High-Throughput and Low-Power Architectures for the AV1 Arithmetic Encoder

34TH SBC/SBMICRO/IEEE/ACM SYMPOSIUM ON INTEGRATED CIRCUITS AND SYSTEMS DESIGN (SBCCI 2021)

Bitencourt, Tulio Pereira; Livi Ramos, Fabio Luis; Bampi, Sergio

被引用次数: 0

2021

A Robust and Power-Efficient Power Line Interference Canceling VLSI Design

34TH SBC/SBMICRO/IEEE/ACM SYMPOSIUM ON INTEGRATED CIRCUITS AND SYSTEMS DESIGN (SBCCI 2021)

da Rosa, Morgana; da Costa, Patricia; da Costa, Eduardo; Almeida, Sergio; Paim, Guilherme; Bampi, Sergio

被引用次数: 0

2021

Configurable Approximate Hardware Accelerator to Compute SATD and SAD Metrics for Low Power All-Intra High Efficiency Video Coding

34TH SBC/SBMICRO/IEEE/ACM SYMPOSIUM ON INTEGRATED CIRCUITS AND SYSTEMS DESIGN (SBCCI 2021)

Lima, Victor H. S.; Stigger, Matheus F.; Soares, Leonardo B.; Diniz, Claudio M.; Bampi, Sergio

被引用次数: 1

2021

Exploring high-order adder compressors for power reduction in sum of absolute differences architectures for real-time UHD video encoding

JOURNAL OF REAL-TIME IMAGE PROCESSING

Paim, Guilherme; Santana, Gustavo M.; Abreu, Brunno A.; Rocha, Leandro M. G.; Grellert, Mateus; da Costa, Eduardo A. C.; Bampi, Sergio

被引用次数: 6

2020

A Cross-Layer Gate-Level-to-Application Co-Simulation for Design Space Exploration of Approximate Circuits in HEVC Video Encoders

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Paim, Guilherme; Rocha, Leandro Mateus Giacomini; Amrouch, Hussam; da Costa, Eduardo Antonio Cesar; Bampi, Sergio; Henkel, Jorg

被引用次数: 33

2020

Residual Syntax Elements Analysis and Design Targeting High-Throughput HEVC CABAC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ramos, Fabio Luis Livi; Saggiorato, Alessandro Via Piana; Zatt, Bruno; Porto, Marcelo; Bampi, Sergio

被引用次数: 15

2020

Local Variability Evaluation on Effective Channel Length Extracted With Shift-and-Ratio Method

IEEE TRANSACTIONS ON ELECTRON DEVICES

Martinez Brito, Juan Pablo; Bampi, Sergio

被引用次数: 0

2020

Two Transistors Voltage-Measurement-Based Test Structure for Fast MOSFET Device Mismatch Characterization

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Brito, Juan Pablo Martinez; Bampi, Sergio

被引用次数: 1

2020

Dynamic and Static Calibration of Ultra-Low-Voltage, Digital-Based Operational Transconductance Amplifiers

ELECTRONICS

Toledo, Pedro; Crovetto, Paolo; Klimach, Hamilton; Bampi, Sergio

被引用次数: 13

2020

An Energy-Efficient and Approximate Accelerator Design for Real-Time Canny Edge Detection

CIRCUITS SYSTEMS AND SIGNAL PROCESSING

Soares, Leonardo Bandeira; Oliveira, Julio; da Costa, Eduardo Antonio Cesar; Bampi, Sergio

被引用次数: 6

2020

Power-Efficient Approximate Newton-Raphson Integer Divider Applied to NLMS Adaptive Filter for High-Quality Interference Cancelling

CIRCUITS SYSTEMS AND SIGNAL PROCESSING

Guidotti, Vagner; Paim, Guilherme; Rocha, Leandro M. G.; Costa, Eduardo; Almeida, Sergio; Bampi, Sergio

被引用次数: 17

2020

Binary CorNET: Accelerator for HR Estimation From Wrist-PPG

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Rocha, Leandro Giacomini; Biswas, Dwaipayan; Verhoef, Bram-Ernst; Bampi, Sergio; Van Hoof, Chris; Konijnenburg, Mario; Verhelst, Marian; Van Helleputte, Nick

被引用次数: 26

2020

Improving devices communication in Industry 4.0 wireless networks

ENGINEERING APPLICATIONS OF ARTIFICIAL INTELLIGENCE

Kunst, Rafael; Avila, Leandro; Binotto, Alecio; Pignaton, Edison; Bampi, Sergio; Rochol, Juergen

被引用次数: 23

2019

Energy-aware cache hierarchy assessment targeting HEVC encoder execution

JOURNAL OF REAL-TIME IMAGE PROCESSING

Monteiro, Eduarda; Grellert, Mateus; Zatt, Bruno; Bampi, Sergio

被引用次数: 0

2019

Hybrid Scratchpad Video Memory Architecture for Energy-Efficient Parallel HEVC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Sampaio, Felipe M.; Zatt, Bruno; Shafique, Muhammad; Henkel, Jorg; Bampi, Sergio

被引用次数: 6

2019

Fast Coding Unit Partition Decision for HEVC Using Support Vector Machines

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Grellert, Mateus; Zatt, Bruno; Bampi, Sergio; da Silva Cruz, Luis A.

被引用次数: 39

2019

Design Methodology to Explore Hybrid Approximate Adders for Energy-Efficient Image and Video Processing Accelerators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Soares, Leonardo Bandeira; Azevedo da Rosa, Morgana Macedo; Diniz, Claudio Machado; Cesar da Costa, Eduardo Antonio; Bampi, Sergio

被引用次数: 52

2019

Power-, Area-, and Compression-Efficient Eight-Point Approximate 2-D Discrete Tchebichef Transform Hardware Design Combining Truncation Pruning and Efficient Transposition Buffers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Paim, Guilherme; Giacomini Rocha, Leandro Mateus; Santana, Gustavo Madeira; Soares, Leonardo Bandeira; Cesar da Costa, Eduardo Antonio; Bampi, Sergio

被引用次数: 20

2019

A 0.3-1.2 V Schottky-Based CMOS ZTC Voltage Reference

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Toledo, Pedro; Cordova, David; Klimach, Hamilton; Bampi, Sergio; Croveti, Paolo S.

被引用次数: 18

2019

A Wideband Low-Noise Variable-Gain Amplifier With a 3.4 dB NF and up to 45 dB Gain Tuning Range in 130-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Baumgratz, Filipe Dias; Saavedra, Carlos; Steyaert, Michiel; Tavernier, Filip; Bampi, Sergio

被引用次数: 19

2019

Improving network resources allocation in smart cities video surveillance

COMPUTER NETWORKS

Kunst, Rafael; Avila, Leandro; Pignaton, Edison; Bampi, Sergio; Rochol, Juergen

被引用次数: 20

2018

A 0.12-0.4 V, Versatile 3-Transistor CMOS Voltage Reference for Ultra-Low Power Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

de Oliveira, Arthur Campos; Cordova, David; Klimach, Hamilton; Bampi, Sergio

被引用次数: 44

2018

40-nm CMOS Wideband High-IF Receiver Using a Modified Charge-Sharing Bandpass Filter to Boost Q-Factor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Baumgratz, Filipe Dias; Ferreira, Sandro Binsfeld; Steyaert, Michiel S. J.; Bampi, Sergio; Tavernier, Filip

被引用次数: 7

2018

Total Dose Effects on Voltage References in 130-nm CMOS Technology

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Colombo, Dalton Martini; Rosseto, Allan; Wirth, Gilson I.; Bampi, Sergio; Goncalvez, Odair Lelis

被引用次数: 5

2018

Exploring approximations in 4-and 8-point DTT hardware architectures for low-power image compression

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Paim, Guilherme; Santana, Gustavo Madeira; Giacomini Rocha, Leandro Mateus; Soares, Leonardo Bandeira; Cesar da Costa, Eduardo Antonio; Bampi, Sergio

被引用次数: 3

2018

A 0.4-3.3 GHz low-noise variable gain amplifier with 35 dB tuning range, 4.9 dB NF, and 40 dBm IIP2

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Baumgratz, Filipe D.; Li, Hao; Tavernier, Filip; Bampi, Sergio; Saavedra, Carlos E.

被引用次数: 9

2018

A High-PSR EMI-Resistant NMOS-Only Voltage Reference Using Zero-VT Active Loads

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Cordova, David; Toledo, Pedro; Klimach, Hamilton; Bampi, Sergio; Fabris, Eric

被引用次数: 7

2017

Complexity control of HEVC encoders targeting real-time constraints

JOURNAL OF REAL-TIME IMAGE PROCESSING

Grellert, Mateus; Zatt, Bruno; Shafique, Muhammad; Bampi, Sergio; Henkel, Joerg

被引用次数: 13

2017

System Design of a 2.75-mW Discrete-Time Superheterodyne Receiver for Bluetooth Low Energy

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Ferreira, Sandro Binsfeld; Kuo, Feng-Wei; Babaie, Masoud; Bampi, Sergio; Staszewski, Robert Bogdan

被引用次数: 14

2017

Picowatt, 0.45-0.6 V Self-Biased Subthreshold CMOS Voltage Reference

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

de Oliveira, Arthur Campos; Cordova, David; Klimach, Hamilton; Bampi, Sergio

被引用次数: 75

2017

Power-Efficient Sum of Absolute Differences Hardware Architecture Using Adder Compressors for Integer Motion Estimation Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Silveira, Bianca; Paim, Guilherme; Abreu, Brunno; Grellert, Mateus; Diniz, Claudio Machado; da Costa, Eduardo Antonio Cesar; Bampi, Sergio

被引用次数: 43

2017

A CMOS low noise transconductance amplifier for 1-6 GHz bands

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Cordova, David; Bampi, Sergio; Fabris, Eric

被引用次数: 2

2016

Design of area and energy-efficient digital CMOS FIR filters with approximate adder circuits

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Soares, Leonardo Bandeira; Cesar da Costa, Eduardo Antonio; Bampi, Sergio

被引用次数: 3

2016

0.75 V supply nanowatt resistorless sub-bandgap curvature-compensated CMOS voltage reference

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Gomez Caicedo, Jhon Alexander; Mattia, Oscar E.; Klimach, Hamilton; Bampi, Sergio

被引用次数: 4

2016

A Reconfigurable Hardware Architecture for Fractional Pixel Interpolation in High Efficiency Video Coding

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Diniz, Claudio M.; Shafique, Muhammad; Bampi, Sergio; Henkel, Joerg

被引用次数: 22

2015

Sub-1 V supply 5 nW 11 ppm/A°C resistorless sub-bandgap voltage reference

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Mattia, Oscar E.; Klimach, Hamilton; Bampi, Sergio

被引用次数: 15

2015

High linearity 24 dB gain wideband inductorless balun low-noise amplifier for IEEE 802.22 band

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Torres Costa, Arthur Liraneto; Klimach, Hamilton; Bampi, Sergio

被引用次数: 7

2015

Sub-1 V band-gap based and MOS threshold-voltage based voltage references in 0.13 μm CMOS

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Colombo, Dalton Martini; Wirth, Gilson; Bampi, Sergio

被引用次数: 11

2015

Special issue: Highlights from the flagship Latin American Symposium on Circuits and Systems: selected papers from LASCAS 2013

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Silva-Cardenas, Carlos; Alarcon, Eduard; Bampi, Sergio

Parallelization of Full Search Motion Estimation Algorithm for Parallel and Distributed Platforms

INTERNATIONAL JOURNAL OF PARALLEL PROGRAMMING

Monteiro, Eduarda; Vizzotto, Bruno; Diniz, Claudio; Maule, Marilena; Zatt, Bruno; Bampi, Sergio

被引用次数: 21

2014

A reduced memory bandwidth and high throughput HDTV motion compensation decoder for H.264/AVC High 4:2:2 profile

JOURNAL OF REAL-TIME IMAGE PROCESSING

Zatt, Bruno; Silva, Leandro M. de L.; Azevedo, Arnaldo; Agostini, Luciano; Susin, Altamiro; Bampi, Sergio

被引用次数: 6

2013

Iterative random search: a new local minima resistant algorithm for motion estimation in high-definition videos

MULTIMEDIA TOOLS AND APPLICATIONS

Porto, Marcelo; Cristani, Cassio; Dall'Oglio, Pargles; Grellert, Mateus; Mattos, Julio; Bampi, Sergio; Agostini, Luciano

被引用次数: 3

2013

Model Predictive Hierarchical Rate Control With Markov Decision Process for Multiview Video Coding

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Vizzotto, Bruno Boessio; Zatt, Bruno; Shafique, Muhammad; Bampi, Sergio; Henkel, Joerg

被引用次数: 5

2013

Synthesis and comparison of low-power high-throughput architectures for SAD calculation

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Walter, Fabio Leandro; Diniz, Claudio Machado; Bampi, Sergio

被引用次数: 11

2012

Power efficient SDS motion estimation architecture using dynamic iteration control and hierarchical adder compressors for real time HDTV video coding

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Porto, Marcelo; Altermann, Joao; Costa, Eduardo; Agostini, Luciano; Bampi, Sergio

被引用次数: 0

2012

Hardware design focusing in the tradeoff cost versus quality for the H.264/AVC fractional motion estimation targeting high definition videos

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Sanchez, Gustavo; Correa, Marcel; Noble, Diego; Porto, Marcelo; Bampi, Sergio; Agostini, Luciano

被引用次数: 5

2012

A DC offset and CMRR analysis in a CMOS 0.35 μm operational transconductance amplifier using Pelgrom's area/accuracy tradeoff

MICROELECTRONICS JOURNAL

Martinez Brito, Juan Pablo; Bampi, Sergio

被引用次数: 3

2009

Design of a digital FM demodulator based on a 2nd-order all-digital phase-locked loop

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Martinez Brito, Juan Pablo; Bampi, Sergio

被引用次数: 6

2008

Analysis and design of amplifiers and comparators in CMOS 0.35 μm technology

MICROELECTRONICS RELIABILITY

Cortes, FP; Fabris, E; Bampi, S

被引用次数: 16

2004

Using BDI agents to improve driver modelling in a commuter scenario

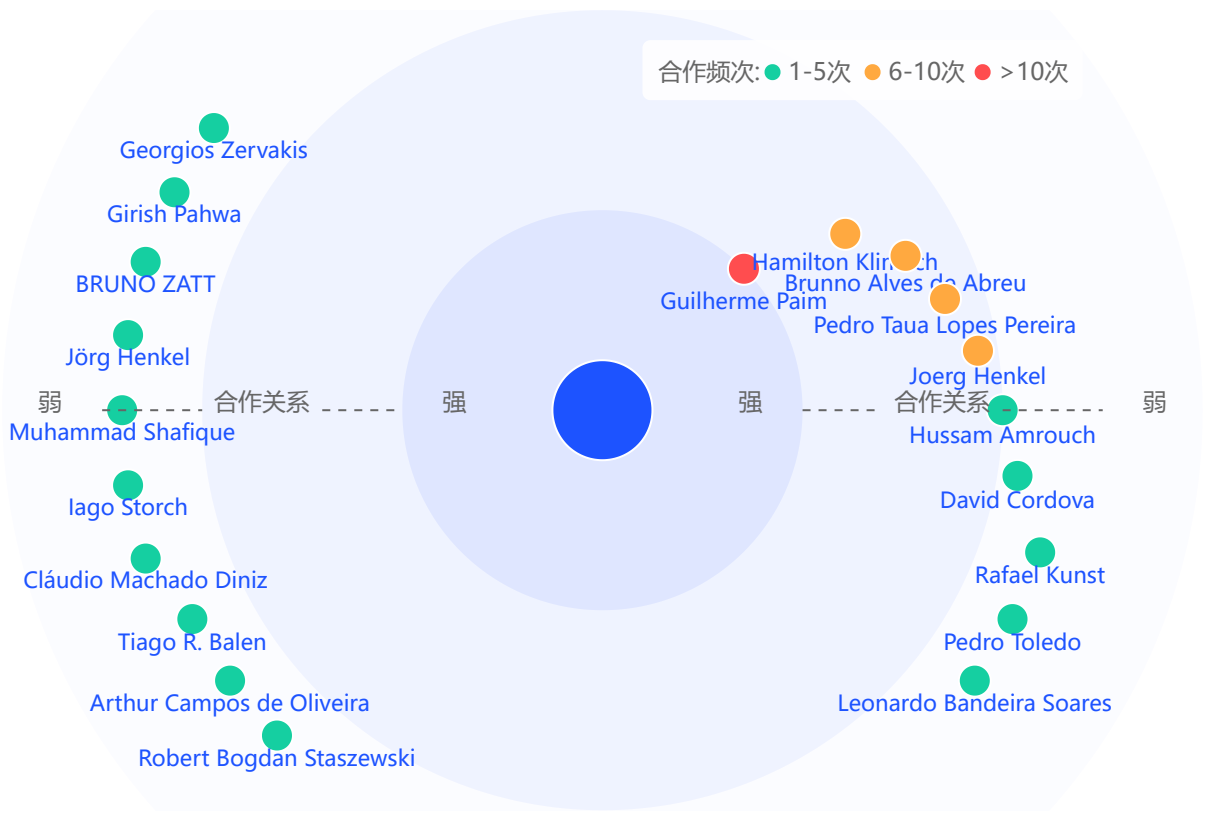
TRANSPORTATION RESEARCH PART C-EMERGING TECHNOLOGIES

Rossetti, RJF; Bordini, RH; Bazzan, ALC; Bampi, S; Liu, R; Van Vliet, D

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2002

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